A Study Comparing NHL Player Salaries Across Positions and Geographic Regions Using Finite Population Sampling

MATH 3430 – Sample Survey Design

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Group project, completed with two teammates

April 6, 2026

This report presents a complete finite-population survey analysis of NHL player salaries using SRSWOR, Stratified Sampling with Proportional Allocation, and Stratified Sampling with Neyman Allocation. Strata are defined by playing position. All computations were performed in R with `set.seed(123)`.

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1 Abstract

This study applies finite population sampling methods to estimate salary characteristics of National Hockey League (NHL) players. The population is $N = 787$ players, each with a recorded pre-tax annual salary and a tax-adjusted after-tax salary, along with playing position (Forward, Defenseman, Goaltender) and team country (Canada or United States). Three sampling designs are evaluated at a fixed total sample size $n = 120$: (1) Simple Random Sampling Without Replacement (SRSWOR), (2) Stratified Sampling with Proportional Allocation, and (3) Stratified Sampling with Neyman Allocation, where strata correspond to playing position. For each design and salary variable, we report the estimated population mean, its variance, standard error, and 95% confidence interval, together with the estimated population total. Relative efficiency comparisons confirm that both stratified designs outperform SRSWOR, with Neyman allocation achieving the lowest variance. A key substantive finding is that tax adjustment substantially reverses the Canada-vs-United States salary ranking: U.S. teams have higher pre-tax average salaries for defensemen and goaltenders, but after taxes Canadian defensemen and forwards retain more, and the gap for goaltenders all but disappears.

Keywords: SRSWOR, stratified sampling, Neyman allocation, proportional allocation, finite population correction, NHL salaries, relative efficiency.

2 Introduction

Salary structures in professional sports leagues are shaped by player performance, market size, contract length, and the tax environment of each team's home jurisdiction. In the NHL, forwards, defensemen, and goaltenders fulfil fundamentally different on-ice roles; whether these positional differences translate into systematic salary gaps is an empirical question. Furthermore, because tax burdens differ between Canadian and U.S. jurisdictions, the pre-tax headline salary may misstate the real economic value of a contract. Without adjusting for taxes, a naive comparison of nominal salaries could lead to incorrect conclusions about which players are better compensated.

This project applies finite population sampling theory to address these questions rigorously. The 787-player dataset is treated as a finite population. Rather than analysing the full population directly, we draw probability samples and use them to construct estimators, thereby demonstrating how estimator precision depends on the chosen sampling design. This is precisely the setting of finite population inference: the population is fixed and finite, randomness enters only through the sampling mechanism, and the goal is to compare designs on the basis of the variance they induce.

Research Objectives

1. Estimate the population mean NHL pre-tax salary μ_{pre} and population mean after-tax salary μ_{after}, together with the corresponding population totals.
2. Quantify estimator precision under three designs: SRSWOR, proportional stratified, and Neyman stratified sampling.
3. Compare designs via relative efficiency (RE).
4. Examine whether mean salary differs across the three playing positions.
5. Assess how tax adjustment alters the Canada vs. United States salary comparison.

3 Population and Data Description

3.1 Population Frame

The sampling frame is the complete set of $N = 787$ NHL players retained after the following cleaning steps:

- Any row with a missing value in Team, Country, Player, Position, SalaryPreTax, or SalaryAfterTax was removed.
- The position code `C` (centre) was recoded to `F` (forward), yielding three mutually exclusive and exhaustive position categories.
- Rows with Position outside $\{F, D, G\}$ or Country outside $\{C, U\}$ were removed.

A derived variable $T_i = Y_i^{\text{pre}} - Y_i^{\text{after}}$ captures the dollar amount lost to taxes for player i.

3.2 Population Composition

Table 1: Population counts by playing position ($N = 787$)

Position	Count (N_h)	Proportion ($W_h = N_h/N$)
Defenseman (D)	269	0.3418
Forward (F)	448	0.5693
Goaltender (G)	70	0.0889
Total	787	1.0000

Table 2: Population counts by country

Country	Count	Proportion
Canada (C)	164	0.2084
United States (U)	623	0.7916
Total	787	1.0000

3.3 Population-Level Salary Summaries

Because the full dataset is available, the true population means serve as benchmarks against which the sample estimates can be evaluated.

Table 3: Population mean pre-tax salary by position and region

Position	Region	Count	Mean Pre-Tax ($)	SD ($)
D	Canada	52	3,545,124	2,757,696
D	United States	217	3,698,438	2,899,817
F	Canada	97	3,805,874	2,887,633
F	United States	351	3,585,417	2,796,670
G	Canada	15	3,195,444	2,658,782
G	United States	55	3,826,136	2,843,115
Overall		787	3,650,480	—

Table 4: Population mean after-tax salary by position and region

Position	Region	Count	Mean After-Tax ($)	SD ($)
D	Canada	52	1,823,065	1,415,306
D	United States	217	1,595,217	1,287,632
F	Canada	97	1,963,960	1,510,241
F	United States	351	1,532,326	1,196,062
G	Canada	15	1,657,259	1,381,689
G	United States	55	1,677,503	1,297,076
Overall		787	1,634,604	—

Table 5: Population mean tax lost ($) by position and region

Position	Region	Count	Mean Tax Lost ($)
D	Canada	52	1,722,059
D	United States	217	2,103,220
F	Canada	97	1,841,914
F	United States	351	2,053,091
G	Canada	15	1,538,186
G	United States	55	2,148,633
Overall		787	2,015,876

The overall population mean pre-tax salary is $3,650,480 and the mean after-tax salary is $1,634,604, implying that the average player loses roughly $2,015,876 – approximately 55% of gross salary – to taxes. Crucially, the regional ranking flips after taxes: for forwards and defensemen, Canadian players retain *more* after-tax income despite having pre-tax salaries that are comparable to, or in some cases lower than, those of their U.S. counterparts.

4 Sampling Design and Methods

4.1 Notation

Let $U = \{1, 2, \ldots, N\}$ be the population of $N = 787$ units and s a probability sample of size $n = 120$. The overall sampling fraction is

$$f = \frac{n}{N} = \frac{120}{787} \approx 0.1525.$$

For stratified designs we index strata by $h \in \{D, F, G\}$, with N_h the stratum population size, n_h the stratum sample size, $W_h = N_h/N$ the stratum weight, and $f_h = n_h/N_h$ the stratum sampling fraction. Let $\bar{Y}_h$ denote the true stratum mean and S_h^2 the true stratum variance, with sample analogues $\bar{y}_h$ and s_h^2.

4.2 Simple Random Sampling Without Replacement (SRSWOR)

Under SRSWOR every subset of size n from U has equal probability of selection. The estimator of the population mean is the sample mean:

$$\hat{\bar{Y}}_{\text{SRS}} = \bar{y} = \frac{1}{n} \sum_{i \in s} y_i.$$

Its estimated variance incorporates the finite population correction (FPC):

$$\widehat{\text{Var}}\left(\hat{\bar{Y}}_{\text{SRS}}\right) = \underbrace{\left(1 - \frac{n}{N}\right)}_{\text{FPC}} \frac{s^2}{n}, \quad \text{where } s^2 = \frac{1}{n-1} \sum_{i \in s} (y_i - \bar{y})^2.$$

The FPC reduces the variance below the with-replacement formula by a factor $(1 - f)$; here $1 - f \approx 0.8475$.

The estimated population total and its standard error are:

$$\hat{T}_{\text{SRS}} = N \, \hat{\bar{Y}}_{\text{SRS}}, \qquad \widehat{\text{SE}}\left(\hat{T}_{\text{SRS}}\right) = N \, \widehat{\text{SE}}\left(\hat{\bar{Y}}_{\text{SRS}}\right).$$

A 95% confidence interval for $\bar{Y}$ is constructed as:

$$\hat{\bar{Y}}_{\text{SRS}} \; \pm \; 1.96 \, \widehat{\text{SE}}\left(\hat{\bar{Y}}_{\text{SRS}}\right).$$

4.3 Stratified Sampling with Proportional Allocation

The population is partitioned into $H = 3$ strata by playing position. Under proportional allocation the stratum sample sizes are

$$n_h^{\text{prop}} = n \cdot W_h = n \cdot \frac{N_h}{N}, \qquad h \in \{D, F, G\}.$$

Since n_h^{prop} is generally non-integer, the *largest-remainder method* is applied: assign $\lfloor n_h^{\text{prop}} \rfloor$ to each stratum, then distribute the remaining $n - \sum_h \lfloor n_h^{\text{prop}} \rfloor$ units one by one to the strata with the largest fractional remainders, subject to $n_h \leq N_h$. The resulting integer allocation is shown in Table 6.

Table 6: Proportional allocation by position

Position	N_h	W_h	$n_h^{\text{raw}} = nW_h$	n_h (integer)
D	269	0.3418	41.02	41
F	448	0.5693	68.32	68
G	70	0.0889	10.67	11
Total	787	1.0000	120.00	120

Within each stratum an independent SRSWOR of size n_h is drawn. The stratified estimator of the population mean is:

$$\hat{\bar{Y}}_{\text{st}} = \sum_{h=1}^{H} W_h \, \bar{y}_h.$$

Its estimated variance is:

$$\widehat{\text{Var}}\left(\hat{\bar{Y}}_{\text{st}}\right) = \sum_{h=1}^{H} W_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{s_h^2}{n_h}.$$

Each term contains a stratum-level FPC $(1 - n_h/N_h)$ and is weighted by W_h^2. The estimated population total is $\hat{T}_{\text{st}} = N \, \hat{\bar{Y}}_{\text{st}}$.

4.4 Stratified Sampling with Neyman Allocation

Neyman (optimal) allocation minimises $\text{Var}(\hat{\bar{Y}}_{\text{st}})$ for a fixed total sample size n:

$$n_h^{\text{Ney}} = n \cdot \frac{N_h S_h}{\sum_{k=1}^{H} N_k S_k}.$$

Strata with larger population size N_h or greater variability S_h receive more sample units. Because the true S_h are unknown in practice, we substitute the population standard deviations computed from the full dataset (oracle allocation). Since S_h differs between pre-tax and after-tax salary, separate Neyman allocations are computed for each variable.

Neyman Allocation for Pre-Tax Salary

Population stratum standard deviations: $S_D = 2{,}868{,}489$, $S_F = 2{,}814{,}813$, $S_G = 2{,}797{,}913$.

$$\begin{aligned}\sum_h N_h S_h &= (269)(2{,}868{,}489) + (448)(2{,}814{,}813) + (70)(2{,}797{,}913)\\ &= 771{,}623{,}582 + 1{,}261{,}036{,}342 + 195{,}853{,}919\\ &= 2{,}228{,}513{,}843.\end{aligned}$$

(The products are computed from the unrounded S_h, so they differ slightly from the product of the rounded values shown.)

$$n_D^{\text{raw}} = 120 \times \frac{771{,}623{,}582}{2{,}228{,}513{,}843} = 41.550,$$

$$n_F^{\text{raw}} = 120 \times \frac{1{,}261{,}036{,}342}{2{,}228{,}513{,}843} = 67.904,$$

$$n_G^{\text{raw}} = 120 \times \frac{195{,}853{,}919}{2{,}228{,}513{,}843} = 10.546.$$

The integer parts (41, 67, 10) sum to 118, so the largest-remainder method assigns the two remaining units to F and D, giving $n_D = 42$, $n_F = 68$, $n_G = 10$.

Table 7: Neyman allocation for pre-tax salary

Position	N_h	S_h	W_h	$N_h S_h$	n_h^{raw}	n_h
D	269	2,868,489	0.342	771,623,582	41.550	42
F	448	2,814,813	0.569	1,261,036,342	67.904	68
G	70	2,797,913	0.089	195,853,919	10.546	10
Total	787			2,228,513,843	120.000	120

Neyman Allocation for After-Tax Salary

Population stratum standard deviations: $S_D = 1{,}313{,}623$, $S_F = 1{,}281{,}268$, $S_G = 1{,}305{,}405$.

$$\begin{aligned}\sum_h N_h S_h &= (269)(1{,}313{,}623) + (448)(1{,}281{,}268) + (70)(1{,}305{,}405)\\ &= 353{,}364{,}587 + 574{,}008{,}064 + 91{,}378{,}350\\ &= 1{,}018{,}751{,}001.\end{aligned}$$

$$n_D^{\text{raw}} = 120 \times \frac{353{,}364{,}587}{1{,}018{,}751{,}001} = 41.623,$$

$$n_F^{\text{raw}} = 120 \times \frac{574{,}008{,}064}{1{,}018{,}751{,}001} = 67.613,$$

$$n_G^{\text{raw}} = 120 \times \frac{91{,}378{,}350}{1{,}018{,}751{,}001} = 10.764.$$

The integer parts (41, 67, 10) again sum to 118; the two remaining units go to G and D, giving $n_D = 42$, $n_F = 67$, $n_G = 11$.

Table 8: Neyman allocation for after-tax salary

Position	N_h	S_h	W_h	$N_h S_h$	n_h^{raw}	n_h
D	269	1,313,623	0.342	353,364,587	41.623	42
F	448	1,281,268	0.569	574,008,064	67.613	67
G	70	1,305,405	0.089	91,378,350	10.764	11
Total	787			1,018,751,001	120.000	120

4.5 Relative Efficiency

The relative efficiency of design A relative to design B is defined as:

$$\text{RE}(A \text{ vs. } B) = \frac{\widehat{\text{Var}}\left(\hat{\bar{Y}}_B\right)}{\widehat{\text{Var}}\left(\hat{\bar{Y}}_A\right)}.$$

A value RE > 1 means design A has smaller estimated variance (is more efficient) than design B.

5 Results

5.1 Pre-Tax Salary: SRSWOR

A SRSWOR of $n = 120$ was drawn with `set.seed(123)`.

$$\hat{\bar{Y}}^{\text{pre}}_{\text{SRS}} = \$4{,}170{,}330, \qquad s^2 = 8{,}925{,}643{,}801{,}435.$$

Estimated variance with FPC:

$$\widehat{\text{Var}}\left(\hat{\bar{Y}}^{\text{pre}}_{\text{SRS}}\right) = \left(1 - \frac{120}{787}\right)\frac{8{,}925{,}643{,}801{,}435}{120} = (0.8475)(74{,}380{,}365{,}012) = 63{,}039{,}013{,}295.$$

$$\widehat{\text{SE}} = \sqrt{63{,}039{,}013{,}295} = \$251{,}076, \qquad 95\%\text{ CI: } (\$3{,}678{,}222,\ \$4{,}662{,}439).$$

$$\hat{T}^{\text{pre}}_{\text{SRS}} = 787 \times 4{,}170{,}330 = \$3{,}282{,}049{,}985, \qquad \widehat{\text{SE}}(\hat{T}) = 787 \times 251{,}076 = \$197{,}596{,}812.$$

5.2 Pre-Tax Salary: Stratified Proportional

$$\hat{\bar{Y}}^{\text{pre}}_{\text{st,prop}} = \$3{,}362{,}231, \qquad \widehat{\text{Var}} = 54{,}651{,}670{,}893, \qquad \widehat{\text{SE}} = \$233{,}777.$$

$$95\%\text{ CI: } (\$2{,}904{,}028,\ \$3{,}820{,}434).$$

$$\hat{T}^{\text{pre}}_{\text{st,prop}} = \$2{,}646{,}076{,}038, \qquad \widehat{\text{SE}}(\hat{T}) = \$183{,}982{,}474.$$

5.3 Pre-Tax Salary: Stratified Neyman

$$\hat{\bar{Y}}^{\text{pre}}_{\text{st,Ney}} = \$3{,}349{,}263, \qquad \widehat{\text{Var}} = 53{,}488{,}449{,}024, \qquad \widehat{\text{SE}} = \$231{,}276.$$

$$95\%\text{ CI: } (\$2{,}895{,}962,\ \$3{,}802{,}563).$$

$$\hat{T}^{\text{pre}}_{\text{st,Ney}} = \$2{,}635{,}869{,}698, \qquad \widehat{\text{SE}}(\hat{T}) = \$182{,}013{,}975.$$

5.4 Pre-Tax Salary Summary

Table 9: Pre-tax salary estimator comparison

Method	$\hat{\bar{Y}}$ (\$)	SE (\$)	CI Lower (\$)	CI Upper (\$)	$\hat{T}$ (\$)
SRSWOR	4,170,330	251,076	3,678,222	4,662,439	3,282,049,985
Proportional	3,362,231	233,777	2,904,028	3,820,434	2,646,076,038
Neyman	3,349,263	231,276	2,895,962	3,802,563	2,635,869,698

Relative efficiencies for pre-tax salary:

$$\text{RE(Prop vs SRS)} = \frac{63{,}039{,}013{,}295}{54{,}651{,}670{,}893} = 1.1535,$$

$$\text{RE(Neyman vs SRS)} = \frac{63{,}039{,}013{,}295}{53{,}488{,}449{,}024} = 1.1786,$$

$$\text{RE(Neyman vs Prop)} = \frac{54{,}651{,}670{,}893}{53{,}488{,}449{,}024} = 1.0217.$$

The variance reduction implied by a relative efficiency is $1 - 1/\text{RE}$. Proportional allocation therefore achieves a 13.3% variance reduction over SRSWOR and Neyman allocation a 15.2% reduction, an incremental 2.1% beyond proportional.

5.5 After-Tax Salary: SRSWOR

The same SRSWOR sample is used for after-tax salary:

$$\hat{\bar{Y}}^{\text{after}}_{\text{SRS}} = \$1{,}899{,}015, \qquad s^2 = 1{,}993{,}664{,}126{,}684.$$

$$\widehat{\text{Var}}\left(\hat{\bar{Y}}^{\text{after}}_{\text{SRS}}\right) = (0.8475)\frac{1{,}993{,}664{,}126{,}684}{120} = 14{,}080{,}622{,}326, \qquad \widehat{\text{SE}} = \$118{,}662.$$

$$95\%\text{ CI: } (\$1{,}666{,}438,\ \$2{,}131{,}592).$$

$$\hat{T}^{\text{after}}_{\text{SRS}} = \$1{,}494{,}525{,}067, \qquad \widehat{\text{SE}}(\hat{T}) = \$93{,}386{,}835.$$

5.6 After-Tax Salary: Stratified Proportional

$$\hat{\bar{Y}}^{\text{after}}_{\text{st,prop}} = \$1{,}540{,}873, \qquad \widehat{\text{Var}} = 12{,}518{,}558{,}006, \qquad \widehat{\text{SE}} = \$111{,}886.$$

$$95\%\text{ CI: } (\$1{,}321{,}575,\ \$1{,}760{,}170).$$

$$\hat{T}^{\text{after}}_{\text{st,prop}} = \$1{,}212{,}666{,}704, \qquad \widehat{\text{SE}}(\hat{T}) = \$88{,}054{,}567.$$

5.7 After-Tax Salary: Stratified Neyman

$$\hat{\bar{Y}}^{\text{after}}_{\text{st,Ney}} = \$1{,}535{,}275, \qquad \widehat{\text{Var}} = 12{,}296{,}975{,}180, \qquad \widehat{\text{SE}} = \$110{,}892.$$

$$95\%\text{ CI: } (\$1{,}317{,}927,\ \$1{,}752{,}623).$$

$$\hat{T}^{\text{after}}_{\text{st,Ney}} = \$1{,}208{,}261{,}463, \qquad \widehat{\text{SE}}(\hat{T}) = \$87{,}271{,}789.$$

5.8 After-Tax Salary Summary

Table 10: After-tax salary estimator comparison

Method	$\hat{\bar{Y}}$ ($)	SE ($)	CI Lower ($)	CI Upper ($)	$\hat{T}$ ($)
SRSWOR	1,899,015	118,662	1,666,438	2,131,592	1,494,525,067
Proportional	1,540,873	111,886	1,321,575	1,760,170	1,212,666,704
Neyman	1,535,275	110,892	1,317,927	1,752,623	1,208,261,463

Relative efficiencies for after-tax salary:

$$\text{RE(Prop vs SRS)} = \frac{14{,}080{,}622{,}326}{12{,}518{,}558{,}006} = 1.1248,$$

$$\text{RE(Neyman vs SRS)} = \frac{14{,}080{,}622{,}326}{12{,}296{,}975{,}180} = 1.1450,$$

$$\text{RE(Neyman vs Prop)} = \frac{12{,}518{,}558{,}006}{12{,}296{,}975{,}180} = 1.0180.$$

5.9 Consolidated Efficiency Comparison

Table 11: Relative efficiency summary for both salary variables

Comparison	Pre-Tax RE	After-Tax RE
Proportional vs. SRSWOR	1.1535	1.1248
Neyman vs. SRSWOR	1.1786	1.1450
Neyman vs. Proportional	1.0217	1.0180

Across both salary variables, stratified sampling consistently improves precision. The incremental gain from Neyman over proportional allocation is small (roughly 2% variance reduction) because the three position strata have nearly identical within-stratum standard deviations. When all S_h are equal, Neyman allocation reduces exactly to proportional allocation, and no further gains are possible from optimal within-stratum weighting.

5.10 Interpretation of Substantive Results

Salary distributions

Both pre-tax and after-tax salary distributions are strongly right-skewed. The majority of players earn near the league minimum ($775,000), while a small number of elite players command contracts up to $17,000,000. This concentration of extreme values drives the large within-stratum standard deviations (around $2,800,000 for pre-tax across all positions) and limits the precision gains achievable through stratification by position alone.

Position comparisons

Mean pre-tax salaries are remarkably similar across positions: \$3,668,801 (D), \$3,633,150 (F), \$3,690,988 (G). The same pattern holds for after-tax salaries. Playing position is therefore a weak predictor of individual salary relative to within-position variability, which explains the modest efficiency gains from stratification.

Regional comparison and the tax reversal

On a pre-tax basis, U.S. teams pay more than Canadian teams for defensemen (\$3,698,438 vs. \$3,545,124) and goaltenders (\$3,826,136 vs. \$3,195,444), while Canadian teams pay more for forwards (\$3,805,874 vs. \$3,585,417). After taxes this picture changes substantially: Canadian defensemen retain \$1,823,065 vs. \$1,595,217 for U.S. defensemen, and Canadian forwards retain \$1,963,960 vs. \$1,532,326 for U.S. forwards. Only for goaltenders does the difference effectively vanish (\$1,657,259 Canada vs. \$1,677,503 United States).

The mean tax lost by U.S. players (\$2,103,220 D, \$2,053,091 F, \$2,148,633 G) consistently exceeds that of Canadian players (\$1,722,059 D, \$1,841,914 F, \$1,538,186 G), confirming that higher U.S. nominal salaries are partly or fully offset by higher tax burdens.

6 Conclusions and Discussion

6.1 Summary of Main Findings

This study estimated NHL salary characteristics from a probability sample of $n = 120$ players drawn from a finite population of $N = 787$.

The true population mean pre-tax salary is \$3,650,480 and the true population mean after-tax salary is \$1,634,604. Both stratified estimators (proportional and Neyman) fell closer to these true values than the SRSWOR estimator, which was inflated by high-salary outliers captured in the particular SRSWOR draw. This illustrates the importance of design: a more precise design is less susceptible to chance imbalance in how outliers are represented.

Stratified sampling with positions as strata improves precision over SRSWOR in all cases. For pre-tax salary, proportional allocation reduces variance by 13.3% and Neyman allocation by 15.2%. For after-tax salary, the figures are 11.1% and 12.7%. The additional gain from Neyman over proportional is approximately 2% in both cases, reflecting near-uniform within-stratum variability across positions.

The most substantive finding is the regional tax reversal. Pre-tax salary comparisons suggest U.S. teams offer higher compensation for defensemen and goaltenders, yet after-tax salary comparisons show that Canadian defensemen and forwards actually retain more income. Any analysis of NHL compensation that relies solely on pre-tax salaries therefore risks misleading conclusions about the financial incentives for players choosing between Canadian and U.S. teams.

6.2 Limitations

1. **Right-skewed distributions.** The high within-stratum variance driven by extreme salaries limits the efficiency gains from stratification by position. Stratifying on contract tier, age, or years of experience might produce more homogeneous strata and larger variance reductions.
2. **Cross-sectional scope.** The dataset reflects a single season. Collective bargaining agreements and roster changes evolve over time; the findings may not generalise to other seasons.
3. **Tax simplification.** The after-tax salary variable is based on simplified tax calculations and does not capture the full complexity of athlete taxation, including state or provincial rates, endorsement income, deferred compensation, or signing bonuses.
4. **Oracle Neyman allocation.** Using full-population standard deviations to set Neyman allocation is an oracle procedure unavailable in a real survey setting, where S_h must be estimated from a pilot sample or auxiliary data. This introduces additional uncertainty not quantified in the present analysis.
5. **Single sample realisation.** All results are based on a single sample draw per design. A simulation study repeating the sampling many times would provide empirical confirmation of the relative efficiency figures.

References

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Appendix A: Complete Annotated R Code

```r
################################################################
# MATH 3430 PROJECT
# NHL Salaries: Pre-Tax and After-Tax Sampling Analysis
# Sheet: "Data Set w Income Tax"
# Methods: SRSWOR, Proportional Stratified, Neyman Stratified
#
# Set the working directory to the repository root, then
# run: source("R/nhl_salary_sampling.R", echo = TRUE)
################################################################

# ---------------------------------------------------------------
# 1. LOAD PACKAGES
# ---------------------------------------------------------------
library(readxl)
library(dplyr)
library(ggplot2)
library(scales)
library(tidyr)
options(scipen = 999)

# ---------------------------------------------------------------
# 2. READ DATA
# ---------------------------------------------------------------
# The workbook is not distributed with this repository.
# See data/README.md for the expected layout.
data_path <- "data/nhl_salaries.xlsx"

df <- read_excel(
  data_path,
  sheet = "Data Set w Income Tax"
)
names(df)[1:6] <- c("Team", "Country", "Player", "Position",
                    "SalaryPreTax", "SalaryAfterTax")

# ---------------------------------------------------------------
# 3. CLEAN DATA
# ---------------------------------------------------------------
df <- df %>%
  mutate(
    Team          = trimws(as.character(Team)),
    Country       = trimws(as.character(Country)),
    Player        = trimws(as.character(Player)),
    Position      = trimws(as.character(Position)),
    SalaryPreTax  = as.numeric(SalaryPreTax),
    SalaryAfterTax = as.numeric(SalaryAfterTax)
  ) %>%
  filter(
    !is.na(Team), !is.na(Country), !is.na(Player),
    !is.na(Position), !is.na(SalaryPreTax), !is.na(SalaryAfterTax)
  ) %>%
  mutate(Position = ifelse(Position == "C", "F", Position)) %>%
  filter(
```

```r
    Position %in% c("F", "D", "G"),
    Country  %in% c("C", "U")
  ) %>%
  mutate(
    Region   = ifelse(Country == "C", "Canada", "United States"),
    TaxLost = SalaryPreTax - SalaryAfterTax
  )

N <- nrow(df)    # 787
n <- 120         # total sample size

# ------------------------------------------------------------
# 4. HELPER: INTEGER ALLOCATION SUMMING TO n
#    Largest-remainder (Hamilton) method
# ------------------------------------------------------------
fix_allocation <- function(raw_alloc, Nh, n_total) {
  alloc_floor <- floor(raw_alloc)
  remainder   <- n_total - sum(alloc_floor)
  frac_part   <- raw_alloc - alloc_floor
  alloc       <- alloc_floor

  if (remainder > 0) {
    add_order <- order(frac_part, decreasing = TRUE)
    i <- 1
    while (remainder > 0 && i <= length(add_order)) {
      idx <- add_order[i]
      if (alloc[idx] < Nh[idx]) {
        alloc[idx] <- alloc[idx] + 1
        remainder  <- remainder - 1
      }
      i <- i + 1
      if (i > length(add_order) && remainder > 0) i <- 1
    }
  }
  if (remainder < 0) {
    sub_order <- order(frac_part, decreasing = FALSE)
    i <- 1
    while (remainder < 0 && i <= length(sub_order)) {
      idx <- sub_order[i]
      if (alloc[idx] > 1) {
        alloc[idx] <- alloc[idx] - 1
        remainder  <- remainder + 1
      }
      i <- i + 1
      if (i > length(sub_order) && remainder < 0) i <- 1
    }
  }
  return(alloc)
}

# ------------------------------------------------------------
# 5. SRSWOR SAMPLE (shared for both salary variables)
# ------------------------------------------------------------
set.seed(123)
```

```r
srs_sample <- df %>% slice_sample(n = n, replace = FALSE)

# ------------------------------------------------------------
# 6. PROPORTIONAL ALLOCATION BY POSITION
# ------------------------------------------------------------
prop_info <- df %>%
  group_by(Position) %>%
  summarise(Nh = n(), .groups = "drop") %>%
  arrange(Position) %>%
  mutate(
    Wh       = Nh / N,
    nh_raw   = n * Nh / N,
    nh_alloc = fix_allocation(nh_raw, Nh, n)
  )
# Result: D=41, F=68, G=11

set.seed(123)
prop_sample_list <- list()
for (i in 1:nrow(prop_info)) {
  h_position <- prop_info$Position[i]
  h_n        <- prop_info$nh_alloc[i]
  prop_sample_list[[i]] <- df %>%
    filter(Position == h_position) %>%
    slice_sample(n = h_n, replace = FALSE)
}
prop_sample <- bind_rows(prop_sample_list)

# ------------------------------------------------------------
# 7. NEYMAN ALLOCATION AND SAMPLING FUNCTION
# ------------------------------------------------------------
draw_neyman_sample <- function(data_full, salary_var,
                               total_n, seed = 123) {
  N_total <- nrow(data_full)
  neyman_info <- data_full %>%
    group_by(Position) %>%
    summarise(
      Nh = n(),
      Sh = sd(.data[[salary_var]]),
      .groups = "drop"
    ) %>%
    arrange(Position) %>%
    mutate(
      Wh       = Nh / N_total,
      NhSh     = Nh * Sh,
      nh_raw   = total_n * NhSh / sum(NhSh),
      nh_alloc = fix_allocation(nh_raw, Nh, total_n)
    )
  set.seed(seed)
  neyman_sample_list <- list()
  for (i in 1:nrow(neyman_info)) {
    h_position <- neyman_info$Position[i]
    h_n        <- neyman_info$nh_alloc[i]
    neyman_sample_list[[i]] <- data_full %>%
      filter(Position == h_position) %>%
```

```r
      slice_sample(n = h_n, replace = FALSE)
  }
  neyman_sample <- bind_rows(neyman_sample_list)
  list(neyman_info = neyman_info, neyman_sample = neyman_sample)
}

# ------------------------------------------------------------
# 8. STRATIFIED ESTIMATOR FUNCTION
# ------------------------------------------------------------
compute_stratified_results <- function(data_full, strat_data,
                                       strata_pop, salary_var) {
  N_total <- nrow(data_full)
  sample_stats <- strat_data %>%
    group_by(Position) %>%
    summarise(
      nh_sample = n(),
      ybar_h    = mean(.data[[salary_var]]),
      s2_h      = var(.data[[salary_var]]),
      .groups   = "drop"
    ) %>%
    arrange(Position)

  stratum_stats <- left_join(strata_pop, sample_stats,
                             by = "Position") %>%
    mutate(
      fh            = nh_sample / Nh,
      Wh_ybarh      = Wh * ybar_h,
      var_component = Wh^2 * (1 - fh) * s2_h / nh_sample
    )

  ybar_st     <- sum(stratum_stats$Wh_ybarh)
  var_ybar_st <- sum(stratum_stats$var_component)
  se_ybar_st  <- sqrt(var_ybar_st)
  t_hat_st    <- N_total * ybar_st
  se_t_st     <- N_total * se_ybar_st

  list(
    ybar_st       = ybar_st,
    var_ybar_st   = var_ybar_st,
    se_ybar_st    = se_ybar_st,
    ci_lower      = ybar_st - 1.96 * se_ybar_st,
    ci_upper      = ybar_st + 1.96 * se_ybar_st,
    t_hat_st      = t_hat_st,
    se_t_st       = se_t_st,
    stratum_stats = stratum_stats
  )
}

# ------------------------------------------------------------
# 9. PRE-TAX SALARY ANALYSIS
# ------------------------------------------------------------
# SRSWOR
ybar_srs_pre <- mean(srs_sample$SalaryPreTax)
s2_srs_pre   <- var(srs_sample$SalaryPreTax)
```

```r
var_srs_pre  <- (1 - n / N) * s2_srs_pre / n
se_srs_pre   <- sqrt(var_srs_pre)
# Results: mean=4170330, SE=251076, CI=(3678222, 4662439)

# Proportional
prop_pre <- compute_stratified_results(df, prop_sample,
               prop_info, "SalaryPreTax")
# Results: mean=3362231, SE=233777, CI=(2904028, 3820434)

# Neyman
ney_draw_pre <- draw_neyman_sample(df, "SalaryPreTax", n)
neyman_pre   <- compute_stratified_results(
                  df, ney_draw_pre$neyman_sample,
                  ney_draw_pre$neyman_info, "SalaryPreTax")
# Results: mean=3349263, SE=231276, CI=(2895962, 3802563)

# Relative efficiency
RE_prop_pre   <- var_srs_pre / prop_pre$var_ybar_st   # 1.1535
RE_neyman_pre <- var_srs_pre / neyman_pre$var_ybar_st # 1.1786

# ------------------------------------------------------------
# 10. AFTER-TAX SALARY ANALYSIS
# ------------------------------------------------------------
# SRSWOR
ybar_srs_after <- mean(srs_sample$SalaryAfterTax)
s2_srs_after   <- var(srs_sample$SalaryAfterTax)
var_srs_after  <- (1 - n / N) * s2_srs_after / n
se_srs_after   <- sqrt(var_srs_after)
# Results: mean=1899015, SE=118662, CI=(1666438, 2131592)

# Proportional
prop_after <- compute_stratified_results(df, prop_sample,
                prop_info, "SalaryAfterTax")
# Results: mean=1540873, SE=111886, CI=(1321575, 1760170)

# Neyman
ney_draw_after <- draw_neyman_sample(df, "SalaryAfterTax", n)
neyman_after   <- compute_stratified_results(
                    df, ney_draw_after$neyman_sample,
                    ney_draw_after$neyman_info, "SalaryAfterTax")
# Results: mean=1535275, SE=110892, CI=(1317927, 1752623)

# Relative efficiency
RE_prop_after   <- var_srs_after / prop_after$var_ybar_st   # 1.1248
RE_neyman_after <- var_srs_after / neyman_after$var_ybar_st # 1.1450

# ------------------------------------------------------------
# 11. POPULATION COMPARISON: PRE-TAX VS AFTER-TAX
# ------------------------------------------------------------
comparison_by_position_region <- df %>%
  group_by(Position, Region) %>%
  summarise(
    Count        = n(),
    MeanPreTax   = mean(SalaryPreTax),
```

```r
    MeanAfterTax = mean(SalaryAfterTax),
    MeanTaxLost  = mean(TaxLost),
    .groups      = "drop"
  )

# ------------------------------------------------------------
# 12. SELECTED VISUALISATIONS
# ------------------------------------------------------------

# Distribution of pre-tax salaries
ggplot(df, aes(x = SalaryPreTax)) +
  geom_histogram(bins = 30, fill = "grey40", color = "black") +
  scale_x_continuous(labels = comma) +
  labs(title = "Distribution of Pre-Tax Salaries",
       x = "Pre-Tax Salary", y = "Frequency")

# Mean salary by position: pre-tax vs after-tax
mean_long <- bind_rows(
  df %>% group_by(Position) %>%
    summarise(MeanSalary = mean(SalaryPreTax), .groups = "drop") %>%
    mutate(Type = "Pre-Tax"),
  df %>% group_by(Position) %>%
    summarise(MeanSalary = mean(SalaryAfterTax), .groups = "drop") %>%
    mutate(Type = "After-Tax")
)
ggplot(mean_long, aes(x = Position, y = MeanSalary, fill = Type)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = c("Pre-Tax" = "#1f5cb6", "After-Tax" = "
     grey50")) +
  scale_y_continuous(labels = comma) +
  labs(title = "Mean Salary by Position: Pre-Tax vs After-Tax",
       x = "Position", y = "Mean Salary", fill = "Salary Type")

# SE comparison across methods
se_df <- data.frame(
  Method = rep(c("SRSWOR", "Proportional", "Neyman"), 2),
  SE = c(se_srs_pre,   prop_pre$se_ybar_st,   neyman_pre$se_ybar_st,
         se_srs_after, prop_after$se_ybar_st, neyman_after$se_ybar_st),
  Type = rep(c("Pre-Tax", "After-Tax"), each = 3)
)
ggplot(se_df, aes(x = Method, y = SE, fill = Type)) +
  geom_col(position = "dodge") +
  labs(title = "Comparison of Standard Errors by Sampling Method",
       x = "Method", y = "SE of Estimated Mean", fill = "Salary Type")
```

Listing 1: Complete R script for NHL salary analysis (MATH 3430)